BY ORDER OF THE
SECRETARY OF THE AIR FORCE

DEPARTMENT OF THE AIR FORCE
INSTRUCTION 62-601

30 JUNE 2025

Developmental Engineering

AIRWORTHINESS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing website at **www.e-Publishing.af.mil** for downloading or ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: SAF/AQRE

Certified by: SAF/AQR Service Authority: USSF SAF/SQ (Stephen G. Purdy, Senior Coordinator)

Supersedes: DAFI62-601, 10 June 2022

Pages: 24

This Department of the Air Force Instruction (DAFI) implements Department of the Air Force Policy Directive (DAFPD) 62-6, *Airworthiness.* This publication applies to all civilian employees and uniformed members of the entire Department of the Air Force (DAF), and those with a contractual obligation to abide by the terms of DAF publications. It applies to the Civil Air Patrol (see **paragraph 3.6.2** for exception) and flights by DAF personnel and DAF contractors in foreign-owned air systems. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the office of primary responsibility (OPR) using DAF Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. Supplements to this publication must be sent to the Deputy Assistant Secretary of the Air Force (Science, Technology and Engineering) (SAF/AQR) for review and coordination before publishing. In accordance with the integrated life cycle management chain of authority specified in DAFI 63-101/20-101, *Integrated Life Cycle Management*, mandates to the program managers and chief engineers in the acquisition execution chain are not considered wing-level mandates. Therefore, tiering in accordance with DAFI 90-160, *Publications and Forms Management,* does not apply to those mandates. The authorities to waive wing-, unit-, and delta-level requirements in this publication are identified with a tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See Department of the Air Force Manual (DAFMAN) 90-161, *Publishing Processes and Procedures*,

for a description of the authorities associated with tier numbers. Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority and to the Air Force Technical Airworthiness Authority (TAA). Non-tiered compliance items may be waived by SAF/AQ, Assistant Secretary of the Air Force (Acquisition, Technology, and Logistics), except items in **Chapter 3,** which may be waived by the Technical Airworthiness Authority. Additional guidance on airworthiness process change approval can be found in **paragraph 4.4** of this instruction. Acquisition program offices submit waiver requests via their acquisition execution chain, except for waiver requests to **Chapter 3** that program offices submit directly to the Technical Airworthiness Authority. Non-program offices submit waiver requests via their command chain to the waiver authority.

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include updates to roles, responsibilities, and processes regarding airworthiness assessments and issuance of airworthiness approvals.

Chapter 1—AIRWORTHINESS ASSURANCE **4**

1.1. Purpose. 4

1.2. Applicability. 4

1.3. Airworthiness Assurance. 4

1.4. Airworthiness Authority. 5

Chapter 2—ROLES AND RESPONSIBILITIES **6**

2.1. Assistant Secretary of the Air Force for Acquisition, Technology & Logistics (SAF/AQ). 6

2.2. Assistant Secretary of the Air Force for Space Acquisition and Integration (SAF/SQ). 6

2.3. Department of the Air Force Airworthiness Authority, Commander, Air Force Materiel Command (AFMC/CC). 6

2.4. Air Force Technical Airworthiness Authority (TAA), Director of Engineering and Technical Management/Services, Air Force Life Cycle Management Center (AFLCMC/EN-EZ). 6

2.5. Deputy Chief of Staff for Operations (AF/A3). 7

2.6. Deputy Chief of Staff, Logistics, Installations Engineering and Force Protection (AF/A4). 7

2.7. Chief of Safety (AF/SE). 7

2.8. Deputy Chief of Space Operation for Operations, Cyber, and Nuclear (SF/S3/4/6/7/10 or SF/COO). 7

2.9. Major Command Commanders, Field Command Commanders, and Director, Air National Guard. 8
2.10. Acquisition Program Managers (as defined in DAFI 63-101/20-101). 8
2.11. Science and Technology Program Managers (as referenced in AFI 61-101, Management of Science and Technology), and Chief Development Testers (CDT) or Test Managers (as defined in DoDI 5000.89_DAFI 99-103, Capabilities-Based Test and Evaluation). 9
2.12. Service Acquisition Lead (as defined in DAFI 63-138, Acquisition of Services). . 10
2.13. Chief Engineers (as defined in DAFI 63-101/20-101)..... 10

Chapter 3—AIRWORTHINESS APPROVAL PROCESS 11

3.1. Overview..... 11
3.2. Adaptive Acquisition Framework (AAF) Construct..... 12
3.3. Contractor-Owned Construct. 14
3.4. Experimental Construct. 16
3.5. Additional Use Cases. 16
3.6. Special Situations for Operational Airworthiness. 17

Chapter 4—OTHER ACTIVITIES RELATED TO AIRWORTHINESS ASSURANCE 19

4.1. Leveraging Approvals Issued by Other Airworthiness Authorities. 19
4.2. Delegation. 19
4.3. Airworthiness Approval Rescission. 19
4.4. Changes..... 19

Attachment 1—GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 20

Chapter 1

AIRWORTHINESS ASSURANCE

1.1. Purpose. Department of Defense (DoD) policy requires all air systems owned, leased, operated, used, designed, or modified by DoD to have an airworthiness assessment and approval. **(T-0)** The Technical Airworthiness Authority (TAA), or their delegated technical authority (DTA), performs an airworthiness assessment for the configuration, usage, and operating environment which supports the issuance of an airworthiness approval. If the flight activity has an existing approval from another airworthiness authority recognized by the TAA (e.g., the Federal Aviation Administration or (FAA)), the TAA's assessment begins with an evaluation of existing approvals to gain a full understanding of the certification baseline and the intended configuration, usage, and operating environment. This instruction contains the processes to conduct assessments and obtain and maintain airworthiness approvals of air systems throughout their lifecycle, enabling safe operations throughout the DAF.

1.2. Applicability. Subject to the exceptions identified in **paragraph 3.6**, this instruction applies to all air systems that the DAF owns, leases, operates, uses, designs, or modifies. DAF policy requires that DAF organizations that meet this applicability ensure airworthiness assessments and approvals occur prior to first DAF flight. **(T-1)** For questions on applicability, contact Air Force Life Cycle Management Center, Director of Engineering and Technical Management/Services, (AFLCMC/EN-EZ).

1.3. Airworthiness Assurance. Airworthiness, as defined in Department of Defense Instruction (DoDI) 5030.61, *DoD Airworthiness Policy*, is the property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits. DAF airworthiness provides DAF personnel (service members and civilians) and contractors the appropriate level of safety of flight adapted to DoD-unique requirements. Airworthiness activities encompass the entire lifecycle management of air systems, to include design, production, operation, and maintenance; these activities may be accomplished by DAF, Joint, FAA, or owning-contractor entities.

1.3.1. Design. The DAF ensures air system designs are independently assessed in accordance with airworthiness criteria appropriate for the air system type and intended usage. The DAF, when appropriate, issues airworthiness approvals governing air systems and their technical data. **Paragraph 3.2** provides implementing guidance for complying with the SecAF policy that DAF-designed or -modified air systems will substantially comply with their appropriate airworthiness criteria and therefore pose reduced risk to the warfighter and to the mission. DAF governs design assessments and approvals through the TAA and implementation of this instruction.

1.3.2. Production. For DAF-acquired air systems, DAF ensures manufacturers evaluate and control their production processes to reliably produce each individual air system in accordance with its approved design. The DAF primarily governs this by requiring manufacturers to produce air systems in accordance with industry quality standards and oversight by Defense Contract Management Agency (DCMA) and the program manager in accordance with DAFI 63-145, *Manufacturing and Quality Management*.

1.3.3. Operations. DAF, primarily with 10- and 11-series publications, ensures air systems are operated in accordance with approved design and associated technical data.

1.3.4. Maintenance. DAF, primarily with 20- and 21-series publications and the AF Technical Order System, ensures air systems are maintained in accordance with the approved technical data and program-approved maintenance concept.

1.4. Airworthiness Authority. The DAF Airworthiness Authority oversees airworthiness by relying upon and integrating these DAF design, production, operations, and maintenance airworthiness assurance activities. The SecAF has designated the Commander, Air Force Materiel Command, as the DAF Airworthiness Authority.

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Assistant Secretary of the Air Force for Acquisition, Technology & Logistics (SAF/AQ). Is responsible for DAF-level airworthiness policy and instruction.

2.2. Assistant Secretary of the Air Force for Space Acquisition and Integration (SAF/SQ). Ensures that Space Force acquisition program offices operate or use only those air systems (including, but not limited to, leased or contracted air systems) that have a valid airworthiness approval.

2.3. Department of the Air Force Airworthiness Authority, Commander, Air Force Materiel Command (AFMC/CC).

2.3.1. Coordinates on high airworthiness risks identified during the airworthiness process, prior to risk acceptance.

2.3.2. Provides Command-level support and resource allocation to enable airworthiness policy implementation.

2.4. Air Force Technical Airworthiness Authority (TAA), Director of Engineering and Technical Management/Services, Air Force Life Cycle Management Center (AFLCMC/EN-EZ).

2.4.1. Serves as the independent DAF technical authority on the design of airworthy air systems.

2.4.2. Advises the DAF Airworthiness Authority on airworthiness matters.

2.4.3. Serves as the DAF Airworthiness Authority's subject matter expert to develop DAF-level airworthiness policy.

2.4.4. Determines whether air system modifications are airworthiness related.

2.4.5. Implements, manages, and controls the design activity of airworthiness.

2.4.5.1. Updates, coordinates, publishes, and distributes Military Handbook (MIL-HDBK)-516, *Airworthiness Certification Criteria*. The AF is the preparing activity for this DoD handbook.

2.4.5.2. Ensures air system program offices incorporate airworthy design practices.

2.4.5.3. Updates, coordinates, publishes, and distributes bulletins, advisories, circulars, and directives to provide processes and procedures and specific notifications necessary to assess and maintain the airworthiness of air systems.

2.4.5.4. Publishes DTA guidance, including basic criteria for DTA qualifications. Appoints select individuals with demonstrated airworthiness expertise as DTAs, delegating in writing specified TAA authorities to be executed on the TAA's behalf.

2.4.5.5. Conducts audits to verify adherence to airworthiness policies, instructions, and procedures.

2.4.6. Organizes, trains, and equips the engineering workforce to design and independently assess air systems.

2.4.6.1. Maintains and utilizes the DAF Airworthiness Office (AWO), AFLCMC/EZZ (**USAF.Airworthiness.Office@us.af.mil**). This office serves as the primary interface to the TAA.

2.4.6.2. Develops engineers by providing training, tools, mentoring, resources, and job experiences necessary to develop their technical expertise.

2.4.6.3. Endorses suitably qualified and experienced individuals to serve as subject matter experts to assist in the execution of the design activity of airworthiness. Defines requirements and conditions of endorsement.

2.4.7. Conducts independent airworthiness assessments in accordance with **Chapter 3** of this document.

2.4.8. Issues airworthiness approvals, when appropriate.

2.4.9. Conducts initial civil aircraft operations and public aircraft operations assessments for civil air systems, in collaboration with the contract requiring activity.

2.4.10. Coordinates on risk assessments for serious and high airworthiness risks identified during the airworthiness process, prior to risk acceptance.

2.4.11. Leads, or participates in, assessments of other airworthiness authorities' assurance systems (see **paragraph 3.6**.).

2.4.12. Serves as the DAF representative to the National Airworthiness Council.

2.4.13. Interfaces with other airworthiness authorities to support formulation of agreements, documents, and handbooks, promote process standardization, improve efficiency, and enhance flight safety.

2.5. Deputy Chief of Staff for Operations (AF/A3). Ensures any operational policy and implementing direction relating to airworthiness that AF/A3 issues is consistent with this instruction, to include flight in foreign-owned aircraft guidance (reference **paragraph 3.6.6**.).

2.6. Deputy Chief of Staff, Logistics, Installations Engineering and Force Protection (AF/A4). Designates an AF maintenance policy advisor to ensure implementing direction is consistent with this instruction.

2.7. Chief of Safety (AF/SE).

2.7.1. Works with the DAF Airworthiness Authority to ensure that safety and risk management policies and implementing direction remain consistent with DAF airworthiness policy and instruction.

2.7.2. Collects airworthiness-related flight safety information and makes it available to organizations consistent with safety privilege guidelines.

2.7.3. Provides an annual summary of air system class A mishap reports to the TAA.

2.7.4. When requested by the TAA, participates in airworthiness assessments.

2.8. Deputy Chief of Space Operation for Operations, Cyber, and Nuclear (SF/S3/4/6/7/10 or SF/COO). Coordinates Space Force participation in DAF airworthiness assurance activities, primarily for air vehicles owned, leased, operated, used, or modified by the field commands.

2.9. Major Command Commanders, Field Command Commanders, and Director, Air National Guard.

2.9.1. Ensure major command, field command, and Air National Guard policy and implementing direction are consistent with DAF airworthiness policy and instruction.

2.9.2. Operate or use only those air systems (including, but not limited to, leased or contracted air systems) that have a valid airworthiness approval.

2.9.2.1. Prior to approving DAF military, civilians, or contractors to fly on foreign-owned military air systems as aircrew or passengers, confirm the existence of a valid DoD recognition of the applicable foreign Military Airworthiness Authority or complete an operational airworthiness appraisal in accordance with the process defined in the Under Secretary of Defense for Acquisition and Sustainment (USD(A&S)) Memorandum, *Flight in Foreign-Owned Military Aircraft Implementation Guidance*, December 1, 2020, available from the DAF AWO. The operational airworthiness appraisal approval authority will only be delegated to a general officer or civilian equivalent in the aircrew or passenger's chain of command.

2.9.2.2. Identify an airworthiness oversight strategy for status of operations (public or civil air operations) and ensure contracts for air services or systems include provisions to support airworthiness assessment requirements.

2.9.2.3. Operate in accordance with TAA issued airworthiness approval, to include maintenance and operations for contracted air services.

2.9.2.4. In coordination with the TAA as necessary, ensure contracted civil aircraft operations (CAO) and maintenance are conducted in accordance with the applicable Federal Aviation Administration Airworthiness (FAA-AW) certificate.

2.9.2.5. In cases where a timely formal airworthiness assessment or approval is not feasible, employ operational risk mitigation measures to satisfy military mission requirements.

2.9.3. Define operating environment and mission usage requirements to aircraft program offices for proper airworthiness planning.

2.9.4. Ensure modifications of assigned air systems are authorized by the air system program manager.

2.10. Acquisition Program Managers (as defined in DAFI 63-101/20-101).

2.10.1. Ensure airworthiness and substantiating data requirements are incorporated into program contractual documents.

2.10.2. Ensure air systems are designed or modified in compliance with airworthiness criteria appropriate for the air system type and intended use.

2.10.3. Obtain and maintain FAA type certification for commercial derivative aircraft acquired or modified by the DAF if the primary mission for the air system is the transport of passengers. For all other commercial derivative aircraft acquired or modified by the DAF, obtain and maintain FAA type certification to the maximum extent practical.

2.10.4. Obtain TAA coordination on serious and high risk assessments, and DAF Airworthiness Authority coordination on high risk assessments, identified during the airworthiness process, prior to seeking risk acceptance. Obtain documented acceptance of all risks by the appropriate authority, in accordance with DAFI 91-202, *The Department of the Air Force Mishap Prevention Program,* prior to exposing people, equipment, or the environment to known hazards. Provide proof of risk acceptance to the TAA.

2.10.5. Notify the TAA, DA, and the lead and using commands when reasonably confident that a hazard has become a serious or high risk. Notify the DAF AWO within 24 hours of a class A mishap determination.

2.10.6. Ensure appropriate airworthiness approvals are obtained prior to first DAF flight.

2.10.7. Ensure manufacturers evaluate and control production processes to reliably produce each individual air system in accordance with its approved design.

2.10.8. Issue Military Certificates of Airworthiness for individual DAF air systems conforming to the configuration identified in the airworthiness approval and in a condition for safe operation.

2.10.8.1. Establish and execute monitoring and surveillance processes for fielded systems to identify when fielded air systems do not conform to a TAA-approved configuration or are no longer in a condition for safe operation.

2.10.8.2. Immediately implement risk mitigation activities or revoke the Military Certificate of Airworthiness of any air system found not to be in a condition for safe operation. Notify the TAA of any revocation of a Military Certificate of Airworthiness.

2.10.9. Interface with the TAA to understand the technical feasibility of conducting planned operations using civil air systems as civil aircraft operations or public aircraft operations. Ensure contracted civil aircraft operations and maintenance are conducted in accordance with the applicable FAA-AW certificate. When determining appropriate operating status, consider potential program impacts such as requirements, oversight responsibilities, cost, and schedule. A legal review is required prior to designation of public aircraft operations to ensure statutory eligibility (see **paragraph 3.3** and Title 49 United States Code § 40102(a)(41) and § 40125).

2.10.10. Incorporate hazards identified during the airworthiness process into system safety programs.

2.10.11. Maintain and manage air system data and configurations in support of life cycle airworthiness.

2.10.12. Ensure that air systems maintain airworthiness upon removal of temporary modifications and are returned to an approved configuration.

2.11. Science and Technology Program Managers (as referenced in AFI 61-101, *Management of Science and Technology*), and Chief Development Testers (CDT) or Test Managers (as defined in DoDI 5000.89_DAFI 99-103, *Capabilities-Based Test and Evaluation*). Will implement the appropriate constructs from **Chapter 3** in accordance with any additional Center-level guidance.

2.12. Service Acquisition Lead (as defined in DAFI 63-138,*Acquisition of Services*). Will implement the contractor owned construct from **Chapter 3**, **paragraph 3.3** and in accordance with DAF-level policy.

2.13. Chief Engineers (as defined in DAFI 63-101/20-101).

2.13.1. Develop and obtain TAA approval of airworthiness plans, certification basis documents, and compliance reports, as required.

2.13.2. When leveraging approvals from other airworthiness authorities (see **paragraph 4.1**), manage the integration of subject approvals with the intended DAF unique configuration, usage, and operating environment to ensure a comprehensive airworthiness assessment.

2.13.3. Provide flight test airworthiness hazards, associated risks, and proof of risk acceptance to the lead developmental test organization. Coordinate with the chief developmental tester or test manager to identify and document conditions and operating limitations appropriate for the test environment.

Chapter 3

AIRWORTHINESS APPROVAL PROCESS

3.1. Overview. In accordance with **paragraph 1.1** and **paragraph1.2**, all air systems owned, leased, operated, used, designed, or modified by the DAF are required to have an airworthiness assessment and approval prior to first DAF flight. This chapter describes the process that any DAF organization that meets the applicability statement in **paragraph 1.2** uses to obtain and document that assessment and approval. The process has four universal steps that apply to all DAF flight activities: planning, assessment, risk acceptance, and approval. Since DAF flight activities occur across a broad spectrum – from a single flight of an experimental aircraft, to the repeated engagement of contracted air services, to the development of a fleet of new military aircraft for formal inclusion in the DAF inventory – the rigor, complexity, and resources required to complete these four universal airworthiness assessment and approval steps will be different. To account for how the universal airworthiness steps are executed in different situations, this chapter identifies in **paragraph 3.2**., **paragraph 3.3**, and **paragraph 3.4** three constructs for executing airworthiness: the Adaptive Acquisition Framework (AAF), Contractor Owned (CO), and Experimental constructs. These constructs accommodate the majority of air systems used by the DAF. For air systems and use cases not explicitly captured in these constructs (e.g., contractor owned government operated (COGO), government owned contractor operated (GOCO)), consult the TAA. In addition, the TAA may tailor and approve changes to any of the processes and constructs in this chapter to address unique needs. In cases where a timely formal airworthiness assessment or approval is not feasible, military commanders retain the prerogative to conduct urgent operational military missions while employing prudent risk mitigation measures in accordance with **paragraph 2.9** above and complying with the waiver approval processes detailed within the opening paragraph.

3.1.1. Airworthiness Planning. Airworthiness planning is executed by the DAF organization that uses, operates, designs, leases or modifies an air system, working with the TAA. The purpose of up-front airworthiness planning is to ensure that the DAF organization is prepared and resourced to apply the tenets and procedures of airworthiness appropriately tailored to the planned air system or air system use and to obtain concurrence from the TAA on the proposed approach. The DAF organization must formally document its airworthiness planning. **(T-1)** The opening paragraph of all airworthiness planning documents must outline which tailored, TAA-approved airworthiness approach the organization is using. **(T-1)** Additional documentation details can be found in applicable airworthiness bulletins.

3.1.2. Assessment. Air systems are assessed in accordance with airworthiness criteria appropriate for the air system type and intended usage and environment. In all cases, airworthiness assessments are documented. However, the scope, methodology, timing, and frequency of airworthiness assessments are different for each construct. The intent of the airworthiness assessment is to obtain a requisite understanding of the system and to identify and characterize risks that must be accepted by the DAF. In addition, the assessment either confirms that a valid airworthiness approval is already in-place (e.g., in the case of an FAA-approved air system remaining in civil status) or provides the basis for a DAF airworthiness approval.

3.1.3. Acceptance. If any risks are identified and characterized in the airworthiness assessment, an appropriate DAF authority decides whether or not to accept those risks. Risk acceptance authorities for all situations are defined in DAFI 91-202 or its supplements.

3.1.4. Airworthiness Approval. Airworthiness approvals authorize system operation within the confines of the approval and are required prior to first flight. The approving entity is the TAA except for contractor owned civil aircraft operations.

3.1.5. In addition to the top-level directive guidance in this chapter, the TAA publishes airworthiness bulletins and circulars on the Airworthiness Center of Excellence sharepoint site at **https://usaf.dps.mil/teams/23230/Airworthiness/**.

3.2. Adaptive Acquisition Framework (AAF) Construct. Air system configurations for which the intent is formal delivery, inspection, and acceptance into the DAF fleet (as commonly captured in DD Form 250, *Material Inspection and Receiving Report*) follow the AAF construct. This construct is applicable for air system configurations resulting from new developmental programs as well as design modification programs executed in accordance with the guidance in DAFI 63-101/20-101. Program managers for major capability programs executed in accordance with the guidance in DoDI 5000.85_DAFI 63-151, *Major Capability Acquisition*, and rapid fielding programs executed in accordance with the guidance in DoDI 5000.80_DAFI 63-146, *Operation of the Middle Tier of Acquisition (MTA)*, must meet the airworthiness assessment and approval requirements of the AAF construct. **(T-1)** Rapid prototyping programs under DoDI 5000.80_DAFI 63-146 may initially elect to follow the experimental construct (**paragraph 3.4**.). However, if a rapid prototyping program transitions to a new acquisition program, the new acquisition program must meet the airworthiness requirements in the AAF construct. Similarly, an urgent capability program executed in accordance with DoDI 5000.81_DAFI 63-147, *Urgent Capability Acquisition*, may initially elect to follow the experimental construct (**paragraph 3.4**), but if the final disposition determination directs transition to a program of record, the new program must comply with the requirements of the AAF construct. Therefore, where practicable, air system rapid prototyping and urgent capability programs should "look ahead" to the possibility that they will transition to formal acquisition programs and need to comply with the requirements of this paragraph.

3.2.1. Fielding Lower-risk Systems to the Warfighter. DAFPD 62-6 establishes the policy expectation that, prior to fielding, DAF-designed or -modified air systems substantially comply with their appropriate airworthiness criteria and therefore pose reduced mishap risk to the warfighter and to the mission. To meet this expectation, new air systems (defined as programs that reach Milestone C/fielding decision after July 2025) must meet the TAA's requirements for award of a Military Type Certificate (MTC) prior to the full-rate production or full-deployment decision point. **(T-1)** In addition, air system modifications must also meet the TAA's requirements for award of MTC. **(T-1)**

3.2.2. Airworthiness Planning. Planning for airworthiness certification should be done prior to source solicitation and account for all phases of the system lifecycle. Additional details can be found in applicable airworthiness bulletins

3.2.3. Certification Basis Development and Approval. The TAA, through the use of subject matter experts, reviews and approves the certification basis. This approval should be obtained prior to System Requirements Review. The procuring entity (e.g., Chief Engineer) must develop a certification basis, which includes: **(T-2)**

3.2.3.1. A current detailed system description and its intended usage.

3.2.3.2. Identification of portions of air system configuration and operations covered by non-AF airworthiness approvals or certifications.

3.2.3.3. Identification of portions of air system configuration and operations covered by AF airworthiness approvals.

3.2.3.4. The applicable standards and methods of compliance from Military Handbook (MIL-HDBK)-516, *Airworthiness Certification Criteria*. These can be tailored to meet program needs, so long as the tailoring provides an appropriate level of safety and is approved by the TAA.

3.2.4. Airworthiness During Test and Evaluation (T&E). High or serious risks and hazards associated with non-compliances to MIL-HDBK-516 criteria are expected during developmental test and evaluation before flight test data are obtained. During test and evaluation, the program manager works with engineering, system safety, and the test communities to continuously reduce risk to the greatest extent practical with the goal of achieving compliance at the end of the flight test program.

3.2.4.1. First Flight Compliance Assessment. Program managers execute a TAA-approved assessment prior to first flight and ensure it is completed in accordance with the approved certification basis. **(T-1)** Risks and hazards are developed for each area of non-compliance. The first flight compliance assessment, to include risks and hazards, is approved by the TAA. Conditions regarding the boundaries of a major revision to the air system configuration and the subsequent requirement for a new first flight assessment should be discussed prior to issuance of the airworthiness approval and be included in the airworthiness plan.

3.2.4.2. Risk Acceptance During T&E. Risk acceptance will be in accordance with DAFI 91-202 or its supplements. **(T-1)** The program manager must notify the TAA and the appropriate decision authority (as defined in DAFI 91-202) of any new serious or high risk identified within the test and evaluation phases. **(T-1)** This is an information only notification and does not drive additional risk acceptance or approval requirements. During the T&E campaign, program managers will utilize existing test and safety review board procedures for assessing risk, bounding operations accordingly, and informing the appropriate risk acceptance approval process within Air Force Test Center (AFTC) or AFMC. **(T-1)** The test and evaluation phase is permitted to continue unless otherwise directed by the decision authority.

3.2.4.3. First Flight Airworthiness Approval. The intent of this policy is a singular airworthiness approval applicable for the duration of the test campaign or other decision-authority-defined scope. However, the TAA must reassess and approve major revisions to the air system configuration within the test campaign. **(T-2)**

3.2.4.3.1. At least annually, the program manager must provide an update on risk posture and high risk resolution to the TAA, Airworthiness Authority (AFMC/CC), and appropriate decision authority (as defined in DAFI 63-101/20-101). **(T-1)**

3.2.4.3.2. The understanding of system performance and artifacts generated during test and evaluation phase are critical to the issuance of an airworthiness approval prior to

full-rate production or full-deployment decision (or fielding decision for a modification). The program manager must identify a method to ensure this information is captured and readily available to support future assessments. **(T-2)**

3.2.5. Capstone Compliance Assessment. After the program manager has reduced risks during T&E, and in support of fielding the system or modification, the TAA will execute a capstone compliance assessment and assess risks of any non-compliances. **(T-1)** The compliance assessment, to include risks and hazards, is approved by the TAA.

3.2.6. Risk Coordination and Decisions in Support of MTC. The program manager will coordinate with the using command to determine if proposed mitigating actions to address airworthiness hazards will result in unacceptable impacts to the planned operations and missions. **(T-1)** The program manager must provide proof of risk acceptance to the TAA. **(T-2)** Risk acceptance will be in accordance with DAFI 91-202.

3.2.7. Capstone Airworthiness Approval. An operational airworthiness approval is required prior to full-rate production or full-deployment decision (or fielding decision for a modification).

3.2.7.1. The TAA issues the airworthiness approval based on the airworthiness assessment, operational and maintenance technical data, and risk acceptance. The airworthiness approval may identify operating limitations which mitigate risk.

3.2.7.2. The program manager must provide final operating limitations to the operator by including them in the operational and maintenance technical data or by other standard means. **(T-2)**

3.2.8. Production Airworthiness.

3.2.8.1. The program manager must issue a Military Certificate of Airworthiness to individual DAF air systems conforming to the configuration identified in the airworthiness approval and in a condition for safe operation. **(T-2)** The program manager must identify the service life limit, as defined on the airworthiness approval, on the Military Certificate of Airworthiness. **(T-2)** The program manager retains the original Military Certificate of Airworthiness in the program office files.

3.2.8.2. DAF organizations that own and operate DAF aircraft must maintain a copy of the Military Certificate of Airworthiness in the binder that stores the air vehicle's Air Force Technical Order Form 781-series, *Aircrew Resource Management System (ARMS) Aircrew/Mission Flight Data Document,* information. **(T-1)**

3.2.8.3. The Military Certificate of Airworthiness remains in effect for the approved service life as long as the air system conforms to the TAA-approved configuration and is in a condition for safe operation (i.e., properly maintained in accordance with approved maintenance documentation, and the system is operated in accordance with the approved flight manual and within the approved mission usage).

3.3. Contractor-Owned Construct. Contractor-owned, contractor-operated (COCO) air systems are owned and operated by a private entity receiving compensation via DAF contract, agreement, or other means to provide products or services in support of research and development activity, science and technology activity, flight testing, training, operational missions, or other DAF interests. COCO air systems used by the DAF typically remain in civil status (called "civil

aircraft operations" – CAO), but under certain circumstances COCO air systems can operate as public aircraft (called "public aircraft operations" – PAO). The requirements for CAO and PAO are statutory (Title 49 United States Code § 40102(a)(41) and § 40125) and implemented in this instruction and Chapter 4 of DAFI 10-3601, *Procedures for Contracted Air Operations for Training Support*. While not binding, FAA Advisory Circular 00-1.1B, *Public Aircraft Operations – Manned and Unmanned,* provides additional information to assist with determining whether operations are eligible for CAO and/or PAO. In all cases, the contract requiring activity should consult legal advisors to ensure compliance with the statutes prior to pursuing a public aircraft designation. In cases where the operation can be legally flown as CAO or PAO, the contract requiring activity will make the decision based on programmatic needs. However, the contract requiring activity should utilize CAO to the maximum extent practicable.

3.3.1. Airworthiness Planning. For COCO air systems, DAF airworthiness planning focuses on suitability of conducting the planned contractor flights as CAO or PAO. The contract requiring activity must document its airworthiness planning. **(T-1)** However, the detail and coordination/approval of the airworthiness planning document is different for CAO and PAO.

3.3.1.1. If the contract requiring activity plans for COCO flights to operate exclusively in CAO, the planning document can be short in accordance with applicable airworthiness bulletins.

3.3.1.2. If the contract requiring activity foresees that at least some COCO flights will operate in PAO, the planning document is more in-depth. At a minimum, the document must identify DAF roles and responsibilities for providing airworthiness oversight of PAO. The DAF may also become responsible for safety and accident investigations as described in Title 49 United States Code § 1132, *Civil Aircraft Accident Investigations*. Therefore, for PAO, the contract requiring activity must coordinate the planning document with its legal advisor and safety advisor, and the planning document must be approved by the TAA. **(T-1)**

3.3.2. Airworthiness assessment. For COCO air systems, the airworthiness assessment focuses on the differences between the air systems' existing FAA design assessments and approvals and the intended use of the air systems by the DAF. If the intended use of a COCO air system by the DAF does not conform to existing FAA airworthiness-approved configuration, usage, and operating environment, the contract requiring activity must pursue alternative approaches (e.g., request contractor work with FAA to expand the civil certificate, alternative status declarations, organic or alternative air systems). Airworthiness assessment documentation details are outlined in applicable airworthiness bulletins.

3.3.3. Risk acceptance. Any DAF risks identified through airworthiness assessments must be accepted by the appropriate authorities within the chain of command of the contract requiring activity in accordance with DAFI 10-3601 and/or DAFI 91-202 or its supplements. **(T-1)** For COCO aircraft, this risk acceptance constitutes an acknowledgement, by the contract requiring activity's chain of command, of DAF operational and mission risks (if any) associated with the DAF's use of the COCO air system. It is not an acceptance or assumption of the contractor's risk and should not disturb liability arrangements in the contract. DAF organizations should consider consulting with legal advisors, safety advisors, and the DAF AWO to develop and document the risk acceptance.

3.3.4. Approval. COCO air systems under CAO operate in accordance with applicable operating regulations under Title 14 Code of Federal Regulations. For CAO, the DAF will use existing FAA airworthiness certificate for the COCO air system as evidence of the required airworthiness approval, assuming the planned, Air Force-directed activity remains within the planned, Air Force directed limitations established in the FAA certificate. For PAO, airworthiness approval must be issued, based upon risk acceptance, by the TAA. **(T-0)**

3.3.5. Documentation. For COCO air systems that will operate exclusively in CAO, DAF contract requiring activities must document the airworthiness planning, assessment, and FAA approvals. For CAO, formal DAF-issued airworthiness approval is not required. Because the DAF assumes greater responsibility for providing oversight of COCO aircraft under PAO, contract requiring activities must obtain a DAF airworthiness approval from the TAA. See the COCO airworthiness guidance on the Airworthiness Center of Excellence sharepoint site for additional resources (**https://usaf.dps.mil/teams/23230/Airworthiness/**).

3.4. Experimental Construct. Experimentation, as used in this publication, refers to primary scientific research and development for the DAF, to include discovery, development, and integration of affordable warfighting technologies for our air, space, and cyberspace force. Experimentation activities can include discovering new technology of DAF relevance, identifying solutions to establish DAF mission gaps, maturing emerging technology into DAF systems, and innovative responses to urgent needs. Delegations for assessment, acceptance, and approval activities to the greatest extent practical are required to meet mission needs and ensure an adequate level of assessment is completed for the required activity.

3.4.1. Assessment Approaches. Assessment approaches should consider duration of flight activity or number of sorties required to achieve program goals; availability of design information; and ability to defer risk mitigation to operational risk management. Assessments may include non-standard approaches, as approved by the TAA. Furthermore, mitigations for residual risk may include operational risk management and operational restrictions.

3.4.2. Once the assessment is completed, the risk acceptance authority as defined in DAFI 91-202 will, where appropriate, accept risks not mitigated through other risk reduction activities (e.g., Test Review Board, Safety Review Board). **(T-1)** The TAA will issue approval based upon risk acceptance. **(T-1)**

3.5. Additional Use Cases. There will be times when an activity does not fit into a particular construct cleanly. Consultation with the AWO will be required to find a path to meet certification requirements. These could include, but are not limited to, the following:

3.5.1. Commercial Derivative Aircraft. Civil air systems acquired or modified by the military are commonly referred to as “commercial derivative aircraft.” Program managers must obtain and maintain FAA type certification for commercial derivative aircraft acquired or modified by the DAF if the primary mission for the air system is the transport of passengers. **(T-1)** For all other commercial derivative aircraft acquired or modified by the DAF, program managers must obtain and maintain FAA type certification to the maximum extent practical. **(T-1)** For more information on the services and type certification processes the FAA provides in support of DAF commercial derivative aircraft, refer to FAA Order 8110.101, *Type Certification Procedures for Military Commercial Derivative Aircraft*.

3.5.2. Foreign Military Sales (FMS). DAF conducts airworthiness assessments but does not provide FMS partners with airworthiness approvals for FMS air systems. If there is a DAF version of the FMS air system, portions of the already completed certification basis, artifacts, and assessments for the DAF system can be useful in performing the airworthiness assessment and meeting the foreign partner's requirements for certification. In addition, during FMS air system development (i.e., before delivery to the FMS partner), DAF organizations may oversee the testing or operation of the FMS aircraft. In that case, the DAF organization must have an airworthiness approval in accordance with the Contractor Owned construct (**paragraph 3.3**), the Experimental construct (**paragraph 3.4**), or a tailored approach approved by the TAA. FMS planning should account for all of this, and FMS case requirements should help define the appropriate construct. For Security Cooperation and Security Assistance programs, the program manager must coordinate Letters of Offer and Acceptance airworthiness requirements with the Technical Airworthiness Authority. **(T-2)** Contact the DAF AWO to determine the path forward for assessing air system airworthiness prior to induction into the modification process.

3.5.3. Temporary Modifications. The program manager must ensure that air systems maintain airworthiness upon removal of temporary modifications and are returned to an approved configuration. **(T-2)**

3.5.4. One Time Flight. In cases when an air system sustains damage which compromises its airworthiness, the TAA may issue a "one time" flight Military Flight Release for operation of the air system under defined and controlled limitations. The TAA will conduct an airworthiness risk assessment, and the program manager must obtain risk acceptance, prior to issuance of the Military Flight Release. **(T-1)**

3.6. Special Situations for Operational Airworthiness. The following are unique circumstances where a TAA-issued airworthiness approval is not required:

3.6.1. DoD Commercial Transportation. These are civil air systems transporting personnel or cargo for the United States Transportation Command (USTRANSCOM) and have flight safety evaluated by USTRANSCOM. See guidance in DoDI 4500.53, *DoD Commercial Air Transportation Quality and Safety Review Program*.

3.6.2. Civil Air Patrol (CAP) Air Systems. These are air systems that operate under CAP governing regulations including, but not limited to, Civil Air Patrol Regulation (CAPR) 70-1, *Civil Air Patrol Flight Management,* and CAPR 66-1, *Civil Air Patrol Aircraft Maintenance Management*.

3.6.3. AF Aero Club Air Systems. These are morale, welfare, and recreation air systems. See the guidance in DAFI 34-101, *Department of the Air Force Morale, Welfare, and Recreation (MWR) Programs and Use Eligibility*.

3.6.4. Flight in non-DAF DoD Air Systems. Non-DAF DoD air systems possessing a valid airworthiness approval issued in accordance with DoDI 5030.61.

3.6.5. Flight in Other United States Government Air Systems. Other United States government agency air systems operated by or transporting DAF personnel must have flight safety evaluated by that agency in accordance with Federal Aviation Regulation and Title 41, Code of Federal Regulations, Part 102-33, *Management of Government Aircraft*. **(T-0)**

3.6.6. Flight in Foreign-Owned Aircraft. Prior to approving DAF military, civilians, or contractors to fly on foreign-owned aircraft not already approved in accordance with **paragraph 3.6.1** as aircrew or passengers, commanders (in accordance with **paragraph 2.9.2.1**) will contact the DAF AWO to confirm the existence of a valid DoD recognition of the applicable foreign Military Airworthiness Authority or complete an operational airworthiness appraisal in accordance with the process defined in the USD(A&S) Memorandum, *Flight in Foreign-Owned Military Aircraft Implementation Guidance*, December 1, 2020, available from the DAF AWO. **(T-0)**

3.6.6.1. Upon request, the DAF AWO will determine if a valid DoD recognition exists for the duration of intended operations. If the command elects to pursue a recognition assessment, the command is responsible for resourcing the recognition assessment performed by the DAF AWO. If the command elects to complete an operational airworthiness appraisal, the DAF AWO provides the checklist for execution by the command.

3.6.6.2. Foreign-owned military air systems undergoing modification within a Security Cooperation or Security Assistance program are not included in this special situation. See **paragraph 3.5.2**.

Chapter 4

OTHER ACTIVITIES RELATED TO AIRWORTHINESS ASSURANCE

4.1. Leveraging Approvals Issued by Other Airworthiness Authorities. An AF airworthiness approval can and should utilize approvals issued by other airworthiness authorities. If another airworthiness authority approval is available to be leveraged or utilized, it should be at the discretion of the TAA.

4.1.1. Air systems approved by the FAA or other DoD airworthiness authorities may be accepted as airworthy when the planned DAF usage is consistent with the existing approved configuration, usage, and operating environment.

4.1.2. Use of non-FAA and non-DoD airworthiness approvals require prior TAA approval.

4.1.3. When leveraging other approvals, the certification basis contains the applicable criteria, standards, and methods of compliance from MIL-HDBK-516, or other comparable means, only for the DAF unique configuration, usage, and operating environment.

4.2. Delegation. The TAA may delegate specific authorities for executing airworthiness activities to delegated technical authorities. The TAA defines requirements and conditions of delegation in a memorandum for record.

4.3. Airworthiness Approval Rescission. The TAA may rescind an airworthiness approval upon discovery of conditions that may invalidate a previously issued airworthiness approval.

4.4. Changes. The TAA may approve specific changes of the activities herein to suit special need or purpose or if circumstances warrant.

WILLIAM D. BAILEY
Acting Assistant Secretary of the Air Force
(Acquisition, Technology & Logistics)

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

41 C.F.R. Part 102-33, *Management of Government Aircraft*

49 U.S.C. § 1132, *Civil Aircraft Accident Investigations*

49 U.S.C. § 40102(a)(41), *Definitions, Public Aircraft*

49 U.S.C. § 40125, *Qualifications for Public Aircraft Status*, current edition

DoDI 4500.53, *DoD Commercial Air Transportation Quality and Safety Review Program,* 7 May 2021

DoDI 5000.80_DAFI 63-146, *Operation of the Middle Tier of Acquisition (MTA)*, 25 November 2024

DoDI 5000.85_DAFI 63-151, *Major Capability Acquisition*, 04 November 2021

DoDI 5000.89_AFI 99-103, *Capabilities-Based Test and Evaluation,* 9 December 2021

DoDI 5030.61, *DoD Airworthiness Policy,* 3 December 2024

Under Secretary of Defense for Acquisition and Sustainment Memorandum, *Flight in Foreign-Owned Military Aircraft Implementation Guidance*, 1 December 2020

DAFPD 62-6, *USAF Airworthiness,* 11 April 2025

DAFI 10-3601, *Procedures for Contracted Air Operations for Training Support,* 24 September 2021

DAFI 34-101, *Department of the Air Force Morale, Welfare, and Recreation (MWR) Programs and Use Eligibility*, 07 March 2022

DAFI 63-101/20-101, *Integrated Life Cycle Management,* 16 Feb 2024

DAFI 63-138, *Acquisition of Services,* 02 Jan 2024

DAFI 63-145, *Manufacturing and Quality Management,* 24 October 2024

DAFI 90-160, *Publications and Forms Management,* 14 April 2022

DAFI 91-202, *The Department of the Air Force Mishap Prevention Program,* 20 March 2020

DAFMAN 90-161, *Publishing Processes and Procedures,* 18 October 2023

AFI 33-322, *Records Management and Information Governance Program,* 23 Mar 2020

AFI 61-101, *Management of Science and Technology,* 14 March 2013

FAA Advisory Circular 00-1.1B, *Public Aircraft Operations – Manned and Unmanned,* 21 September 2018

FAA Order 8110.101, *Type Certification Procedures for Military Commercial Derivative Aircraft*, 25 February 2015

CAPR 66-1, *Civil Air Patrol Aircraft Maintenance Management*, 14 September 2015

CAPR 70-1, *Civil Air Patrol Flight Management*, 05 August 2018

MIL-HDBK-516, *Airworthiness Certification Criteria*, 12 December 2014

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

AF Technical Order Form 781, *Aircrew Resource Management System (ARMS) Aircrew/Mission Flight Data Document*

Abbreviations and Acronyms

AAF—Adaptive Acquisition Framework

AF—Air Force

AFI—Air Force Instruction

AFLCMC—Air Force Life Cycle Management Center

AFPD—Air Force Policy Directive

AWO—Airworthiness Office

CAO—Civil Aircraft Operations

CAP—Civil Air Patrol

CAPR—Civil Air Patrol Regulation

CO—Contractor Operated

COCO—Contractor Owned Contractor Operated

DA—Decision Authority

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFPD—Department of the Air Force Policy Directive

DCMA—Defense Contract Management Agency

DoD—Department of Defense

DoDI—Department of Defense Instruction

DTA—Delegated Technical Authority

FAA—Federal Aviation Administration

FAA-AW—Federal Aviation Administration Airworthiness

FMS—Foreign Military Sales

MFR—Military Flight Release

MIL-HDBK—Military Handbook

MIL-STD—Military Standard

MTA—Middle Tier Acquisition

MTC—Military Type Certificate

OPR—Office of Primary Responsibility

PAO—Public Aircraft Operations

SAF—Secretariat Offices within the Headquarters Air Force

SecAF—Secretary of the Air Force

T&E—Test and Evaluation

TAA—Technical Airworthiness Authority

USAF—United States Air Force

USD(A&S)—Under Secretary of Defense for Acquisition and Sustainment

USTRANSCOM—United States Transportation Command

Office Symbols

SAF/AQ—Assistant Secretary of the Air Force (Acquisition, Technology, and Logistics)

SAF/SQ—Assistant Secretary of the Air Force for Space Acquisition and Integration

SAF/AQR—Deputy Assistant Secretary of the Air Force (Science, Technology, and Engineering)

SAF/AQRE—Engineering and Workforce Management Division, Office of the Deputy Assistant Secretary of the Air Force (Science, Technology, and Engineering)

AFMC/CC—Department of the Air Force Airworthiness Authority, Commander, Air Force Materiel Command

AFLCMC/EN-EZ—Air Force Lifecycle Management Center, Director of Engineering and technical Management/Services

AF/A3—Deputy Chief of Staff for Operations

AF/A4—Deputy Chief of Staff, Logistics, Installations Engineering and Force Protection

AF/SE—Chief of Safety

AFRL—Air Force Research Laboratory

AFTC—Air Force Test Center

SF/S3/4/6/7/10 or SF/COO—Deputy Chief of Space Operation for Operations, Cyber, and Nuclear

Terms

Air System—An air vehicle plus the training and support systems for the air vehicle (e.g., communications, control, ground or surface control station, launch and recovery, and support elements), and any weapons to be employed on the air vehicle. For example, an unmanned aircraft system is an air system. An air vehicle, manned or unmanned, is a subset of its associated air system. (See MIL-HDBK-516). This definition does not include space systems or space vehicles.

Airworthiness—The property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits.

Airworthiness Approval—Documents issued by an empowered airworthiness authority and may take a number of different forms (e.g., airworthiness release, military-type certificate, flight clearance) depending on specific airworthiness authority policy. An airworthiness approval affirms that the appropriate tenets of the airworthiness process are met and that the aircraft or air system was assessed against the required airworthiness standards and any residual risk to aircrew, ground crew, passengers, or to third parties has been accepted by the appropriate authority.

Airworthiness Authority—An individual who has the legal mandate to develop and enforce pertinent rules, regulations, and policy governing airworthiness.

Certification Basis—A document comprised of a system description, the set of approved airworthiness certification criteria, standards, and methods of compliance that apply to a specific air system design. It is typically derived from MIL-HDBK-516.

Civil Aircraft Operations—All aircraft operations other than those conducted as public aircraft operations in accordance with applicable law.

Compliance Report—A document comprised of a system description, the approved certification basis, references to compliance data, findings of compliance and non-compliance, hazards associated with non-compliance, and risk levels for those hazards.

Delegated Technical Authority (DTA)—Individuals with demonstrated airworthiness expertise that can perform specified authorities on behalf of the TAA as delegated formally in writing

Military Certificates of Airworthiness—The document issued by a program manager to each individual aircraft that provides evidence of conformance to the configuration identified in the airworthiness approval and its condition relative to safe operation.

Military Flight Release—An airworthiness approval for an air system design that does not meet the full standards or intent of a Military Type Certificate.

Military Type Certificate—An airworthiness approval based on evidence that the air system design is substantially in compliance with its approved certification basis (typically only low or medium risks remain due to non-compliance).

Operating Environment—The surroundings or conditions in which an aircraft operates including, but not limited to temperatures, loads, ambient environmental conditions, moisture and fluid exposures, electromagnetic spectrum, radiation, maintenance, and ground handling.

Passenger—An individual (to include service members, DoD civilians, and contractors) onboard the aircraft who is not on the flight authorization as an aircrew member.

Public Aircraft Operations—Public Aircraft Operations is the operation of an aircraft that meets the legal definition of "public aircraft" established in 49 U.S.C. § 40102(a)(41) and the legal qualifications for public aircraft status outlined in 49 U.S.C. § 40125.

Public Aircraft Operations Declaration—Formal written notice of public aircraft operations for designated flights, provided by an authorized government official to the contractor.

Target Risk—The projected risk level the program manager plans to achieve by implementing mitigation measures consistent with the system safety design order of precedence.

Technical Airworthiness Authority (TAA)—The independent AF official authorized to implement, manage and control the design airworthiness element of DAF airworthiness. Authorities include but are not limited to defining airworthiness standards, approving the certification basis, assessing compliance and risks, and issuing airworthiness approval.